

Joemon Jesudas

Gen AI Engineer | ML Engineer | MLOps Engineer

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SKILLS

Primary Skills: Python, SQL, Machine Learning, Generative AI, LLM Application Development, RAG (Retrieval-Augmented Generation), LangChain, Azure OpenAI, Azure AI Foundry, Azure Document Intelligence, PGVector, Vector Databases, Embeddings, Prompt Engineering, FastAPI, MCP Frameworks

Secondary Skills: Azure Cloud Services, Docker, Kubernetes, MLflow, Grafana, PySpark, Apache Airflow, Power BI, Streamlit, DVC, Git, GitHub Actions, Azure Data Lake Storage, Angular.js

PROFESSIONAL EXPERIENCE

Allianz Services

May 2022 - Present

ML/Gen AI Engineer

- Architected and deployed end-to-end RAG pipelines using LangChain, Azure OpenAI, and PGVector, enabling semantic search over 10,000+ insurance policy documents with sub-second retrieval and contextually accurate LLM-generated responses.
- Developed a RAG-based liability prediction system that analyzes accident descriptions and recommends liability decisions for claim handlers, improving decision consistency by 30% and reducing average manual review time.
- Designed and implemented GenAI-powered document extraction pipelines using Azure Document Intelligence, performing structured data extraction, multi-stage validation checkpoints, and automated section-wise summarization using GPT-4 models.
- Engineered chunking strategies, embedding pipelines, and vector indexing workflows using PGVector and OpenAI embeddings to optimize retrieval relevance and reduce hallucination in RAG applications.
- Built custom prompt templates, few-shot examples, and guardrail logic (content filtering, output validation, hallucination checks) for high-accuracy LLM responses across document understanding and claims processing tasks.
- Developed and deployed traditional machine learning models (Gradient Boosting, Random Forest, XGBoost) for insurance claims handling, resulting in savings of more than 5 million GBP for the client.
- Built RESTful APIs using FastAPI to serve ML model predictions and LLM-generated responses, enabling seamless integration with downstream applications and front-end interfaces.
- Built web-based ML and GenAI applications using Streamlit, enabling claim handlers and business stakeholders to interact with model predictions and LLM-generated insights through intuitive interfaces.
- Designed and scheduled cron jobs for automated data extraction from Azure Data Lake, deploying them in Kubernetes clusters for reliable, production-grade data orchestration.
- Implemented model monitoring using MCP and Claude Desktop integrated with Evidently AI to generate natural-language insights and proactively identify data drift or model performance anomalies.
- Developed interactive Grafana dashboards to monitor LLM and ML pipeline health, combining telemetry, token usage, latency metrics, SQL-based aggregations, and operational KPIs.
- Containerized LLM and ML solutions using Docker and orchestrated deployments on Kubernetes via Helm charts, ensuring reproducible, scalable, and environment-consistent production deployments.
- Implemented CI/CD pipelines with GitHub Actions, automating build, test, and deployment workflows for ML/LLM applications, reducing deployment cycle time significantly.
- Managed data versioning and experiment tracking using DVC and MLflow, ensuring reproducibility across model iterations and RAG configuration changes.

Cognizant Technology Solutions

Dec 2018 - May 2020

Programmer Analyst

- Managed SAP security operations including role provisioning and authorization maintenance across multiple business units, ensuring compliance with internal security policies.
- Streamlined customer support processes through ticketing systems, handling over 120+ tickets per month, improving issue resolution efficiency by 40% and boosting overall customer retention rates by 15%.
- Monitored SAP system alerts, background jobs, RFC connections, and system logs, proactively identifying and resolving performance issues.

Orglens

Dec 2021 - March 2022

Social Network Analyst - Internship

- Conducted social network analysis to identify key influencers, community structures, and information flow patterns within organizational networks.
- Created visual representations of social networks using NetworkX to illustrate complex relationships, structures, and communication patterns.
- Integrated data from multiple sources (social media platforms, communication logs) to perform comprehensive network analysis and derive actionable insights.

PUBLICATIONS

- "Model Monitoring With Grafana and Dynatrace: A Comprehensive Framework For Ensuring ML Model Performance" - <http://dx.doi.org/10.21474/IJAR01/18684>

Co-authored with Gopika SB and Vinod Menon

- "Claims Negotiation Amount Predictor: A Gradient Boosting Regression Based Machine Learning Solution with Model Monitoring" - <http://www.ijrar.org/papers/IJRAR23D1678.pdf>

Co-authored with Surendra Prabu, Mohammed Abdullah and Diana Varghese

EDUCATION & ACADEMICS

Masters in Computer Science

2020 - 2022

Rajagiri College of Social Sciences (Autonomous), MG University | CGPA: 3.96/5.0 (B+)

Bachelors in Computer Application

2015 - 2018

Kristu Jyoti College of Management and Technology, MG University | CCPA: 7.62/10.0 (B)

AWARDS AND CERTIFICATIONS

- Certified in Azure AI Engineer Associate - AI102
- Certified in Azure Fundamentals (Microsoft) - AZ900
- Certification in MLOps - Cloudxlabs
- Certification in AIOps - iNeuron
- Allianz Best All-rounder Q2 - 2025
- Allianz Best All-rounder Q1 - 2025
- Allianz Best Performer Award, Feb - 2024
- Allianz Delivery Excellence Champion - 2023
- Allianz Best Performer Award, April - 2023
- Allianz Best Performer Award, July - 2023
- Allianz Best Performer Award, Dec - 2022